L11_S2_A.pdf
PDF

I want you to generate a lecture scribe with the material provided.
the scribe should serve as an exam reference material which includes detailed definitions and annotations, assumptions and conditions, statement of theorems or results, proofs or proof sketches, step by step derivations and worked examples only if they appear in the provided content, with intermediate steps and clear logical flow. Do not introduce new content.

follow the below structure for the scribe to be generated-
Header information like course name, lecture name or number, topic and name of the instructor
Introduction and overview- which includes brief outline of all the topics covered during the lecture
main body content- which includes the above mentioned content requirements
important notes/ remarks
conclusion

Below is the complete lecture scribe generated **strictly from the provided slides** L11_S2_A , following the exact structure requested and without introducing any new content beyond what appears in the material.

Lecture Scribe

Header Information

Course Name: CSE400 – Fundamentals of Probability in Computing

Lecture Number & Title: Lecture 11 – Transformation of Random Variables

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(Source: Slide 1 L11_S2_A)

Introduction and Overview

This lecture covers:

1. **Transformation of Random Variables**
   - Learning transformation techniques for random variables.
   - Determining the PDF of a new random variable obtained via transformation.
2. **Function of Two Random Variables**
   - Joint transformations and derived distributions.
   - Determining the distribution of a function of two RVs.
3. **Illustrative Example**
   - Detailed derivation for the case:

$$Z = X + Y$$

(Outline based on Slide 2 L11_S2_A)

The main objective emphasized in the lecture is:

> **How to find the PDF of a new transformed random variable?**

An assumption explicitly noted:

> The PDF of X, i.e., $f_X(x)$, is known *a priori*.
> (Slide 2 L11_S2_A)

Main Body Content

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1. Transformation of a Single Random Variable

Let:

$$Y = g(X)$$

We assume:

- X is a continuous random variable.
- $f_X(x)$ is known.
- $g(\cdot)$ is monotonic (cases handled separately for increasing and decreasing).

(Slides 3–4 L11_S2_A)

Case 1: $g(x)$ is Monotonically Increasing

Step 1: Compute CDF of Y

$$F_Y(y) = P(Y \leq y)$$

Since $Y = g(X)$,

$$F_Y(y) = P(g(X) \leq y)$$

If g is increasing, then:

$$g(X) \leq y \iff X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

(Slide 3 L11_S2_A)

Step 2: Differentiate to Obtain PDF

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since:

$$\frac{d}{dy} g^{-1}(y) = \frac{dx}{dy}$$

We obtain:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

(Slides 3–4 L11_S2_A)

Case 2: $g(x)$ is Monotonically Decreasing

Step 1: CDF

$$F_Y(y) = P(Y \leq y)$$

Since g is decreasing:

$$g(X) \leq y \iff X \geq g^{-1}(y)$$

Thus:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

(Slide 4 L11_S2_A)

Step 2: Differentiate

Differentiating gives:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

The absolute value accounts for the negative derivative in decreasing transformations.

(Slide 4 L11_S2_A)

Final Result (Single Variable Transformation)

$$\boxed{f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}}$$

(Slide 4 L11_S2_A)

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2. Worked Example (From Slides)

(Slide 5 L11_S2_A)

Given:

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Transformation:

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

Step 1: Find Inverse

$$y = \sin\left(\frac{\pi x}{2}\right)$$

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

Step 2: Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

Step 3: Apply Transformation Formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Since $f_X(x) = \frac{1}{2}$,

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1 - y^2}}$$

$$f_Y(y) = \frac{1}{\pi\sqrt{1 - y^2}}, \quad -1 < y < 1$$

Otherwise:

$$f_Y(y) = 0$$

(Slide 5 L11_S2_A)

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3. Function of Two Random Variables

(Slides 6–7 L11_S2_A)

Let:

$$Z = X + Y$$

We are required to:

1. Find $f_Z(z)$
2. Find $f_Z(z)$ if X and Y are independent
3. If $X \sim N(0,1)$, $Y \sim N(0,1)$, prove $Z \sim N(0,2)$
4. If X and Y are exponential with parameter λ, find $f_Z(z)$

(Slide 6 L11_S2_A)

Derivation Using CDF Method

$$F_Z(z) = P(Z \leq z)$$

$$= P(X + Y \leq z)$$

This corresponds to integrating the joint density over the region:

$$x + y \leq z$$

(Slide 7 L11_S2_A)

Thus,

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x,y)\, dx\, dy$$

Integration may be performed:

- Horizontal strip method
- Vertical strip method

(as illustrated in the geometric diagram on Slide 7 L11_S2_A)

After computing $F_Z(z)$, differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Important Notes / Remarks

1. The PDF of the original variable must be known.
2. Monotonicity of transformation is essential for single-variable transformation.
3. Absolute value of derivative is required in transformation formula.
4. For sum of two random variables:
   - Use CDF approach:

     $$F_Z(z) = P(X + Y \leq z)$$

   - Convert probability region into double integral.
   - Carefully determine limits from geometry.
5. Region of integration must match the inequality $x + y \leq z$.
6. Differentiation of CDF gives the final PDF.

(All based strictly on Slides 2–7 L11_S2_A)

Conclusion

In this lecture, we studied:

- Transformation of a single random variable using the CDF method.
- Derivation of the general transformation formula:

  $$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

- Application to a trigonometric transformation example.
- Extension to functions of two random variables.
- Derivation approach for:

  $$Z = X + Y$$

  using geometric interpretation and double integration.

The lecture establishes systematic techniques to derive PDFs of transformed random variables and lays the foundation for handling joint transformations and sums of random variables.

(Complete content derived strictly from Lecture 11 slides L11_S2_A .)

Sources